

Implementation of ETL and ELT Pipelines for Olist E-Commerce Analytics Using SQLite Data Warehouse and Power BI Dashboard

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Abstract—This paper presents the implementation of ETL and ELT pipelines for an e-commerce analytics project using the Olist Brazilian E-Commerce Dataset and the Nager.Date Public Holiday API. The project builds a SQLite-based data warehouse with star schema design and provides analytical queries and a Power BI dashboard. The ETL pipeline transforms data using Python and Pandas before loading it into the warehouse, while the ELT pipeline loads raw data first and performs transformations using SQL inside the warehouse. The final fact table contains 112,650 rows and supports analysis of revenue, orders, product categories, payment types, customer regions, delivery performance, reviews, and holiday effects. The dashboard visualizes key performance indicators and provides a comparison between ETL and ELT approaches.

Keywords—ETL, ELT, big data, data warehouse, SQLite, Power BI, Olist, API enrichment

I. INTRODUCTION

E-commerce platforms generate transactional data that can be analyzed to support business decision-making. Data from orders, products, payments, customers, reviews, and delivery activities can be transformed into analytical information when organized in a data warehouse. This project implements two different data pipeline approaches, namely Extract, Transform, Load (ETL) and Extract, Load, Transform (ELT), using the same data sources to compare their implementation patterns.

The main objective of this project is to build reproducible ETL and ELT pipelines, create a SQLite data warehouse using a star schema model, provide analytical SQL queries, and visualize the results through a Power BI dashboard. The dashboard focuses on sales and delivery analytics, including revenue trend, product category contribution, payment distribution, customer state contribution, review score, and holiday versus non-holiday revenue.

II. DATA SOURCES

The first data source is the Olist Brazilian E-Commerce Dataset from Kaggle. This dataset contains transaction-related CSV files, including customers, orders, order items, payments, reviews, products, sellers, geolocation, and product category translation. The Olist dataset is used as the primary data source for building the fact table and dimension tables.

The second data source is the Nager.Date Public Holiday API. The API is used to retrieve Brazilian public holiday data for 2016, 2017, and 2018 using the country code BR. The holiday data is used as enrichment data by joining order

dates with holiday dates to generate holiday-related features, such as `is_public_holiday` and `holiday_name`.

III. SYSTEM ARCHITECTURE

The system architecture consists of data sources, raw storage, data lake, ETL pipeline, ELT pipeline, SQLite data warehouse, analytical SQL queries, and Power BI dashboard. In the ETL path, data is extracted and transformed using Python and Pandas before being loaded into the SQLite warehouse. In the ELT path, raw data is loaded first into SQLite raw tables, then transformed using SQL into staging tables and final star schema tables.

The architecture separates raw data, processed data, curated data, warehouse outputs, and dashboard artifacts. This organization supports reproducibility because the notebooks can be rerun to regenerate outputs from the original sources.

IV. ETL PIPELINE IMPLEMENTATION

The ETL pipeline follows the Extract, Transform, Load approach. The extraction stage reads Olist CSV files and retrieves holiday data from the Nager.Date API. Raw data is copied into the raw folder and the data lake raw zone. The transformation stage includes column standardization, datetime conversion, aggregation before join, controlled join, holiday enrichment, feature engineering, missing value handling, duplicate handling, outlier handling, normalization, categorical encoding, and data validation.

The controlled join uses `order_id` and `order_item_id` as the main grain so that one row represents one item in one order. Payments and reviews are aggregated before joining to prevent invalid duplication. The final ETL fact table contains 112,650 rows and 29 columns after being loaded

into SQLite. The ETL warehouse consists of fact_order_items as the fact table and dim_customer, dim_product, dim_seller, dim_payment, and dim_date as dimension tables.

V. ELT PIPELINE IMPLEMENTATION

The ELT pipeline follows the Extract, Load, Transform approach. In this approach, Olist CSV data and holiday API data are extracted and loaded directly into raw tables inside the SQLite warehouse. The raw tables include raw_customers, raw_geolocation, raw_order_items, raw_payments, raw_reviews, raw_orders, raw_products, raw_sellers, raw_category_translation, and raw_holidays.

After raw loading, transformations are performed inside the warehouse using SQL. The ELT process creates staging tables such as stg_payments_agg, stg_reviews_agg, stg_geolocation_agg, stg_products_en, and stg_fact_order_items_enriched. The final ELT star schema consists of fact_order_items_elt, dim_customer_elt, dim_product_elt, dim_seller_elt, dim_payment_elt, and dim_date_elt. The final ELT fact table also contains 112,650 rows.

VI. DATA WAREHOUSE DESIGN

The data warehouse uses SQLite and follows a star schema design. The fact table grain is one row per order item. The fact table stores measures such as price, freight value, payment value, review score, delivery days, delivery delay days, late delivery flag, total item value, freight ratio, product volume, and public holiday flag.

The dimension tables describe business entities related to the fact table. Customer dimension stores customer location information. Product dimension stores product category and physical attributes. Seller dimension stores seller location. Payment dimension stores payment type and installment information. Date dimension stores date attributes, including year, month, day, weekend flag, and holiday flag.

TABLE I. DATA WAREHOUSE OUTPUT SUMMARY

Pipeline	Final Fact Table	Rows
ETL	fact_order_items	112,650
ELT	fact_order_items_elt	112,650

VII. ANALYTICAL QUERIES AND DASHBOARD

Analytical SQL queries are provided to support verification and dashboard development. The ETL analytical queries include monthly revenue, top product categories, payment type distribution, customer state revenue, seller state revenue, late delivery versus review, holiday versus non-holiday, and weekend versus weekday analysis. The ELT analytical queries include monthly revenue, category revenue, payment type distribution, customer state revenue, late delivery review, and holiday versus non-holiday analysis.

The dashboard is created using Power BI Desktop. The main dashboard page displays key performance indicators such as total revenue, total orders, total customers, average review score, and late delivery percentage. It also visualizes revenue trend, top product categories, payment type

distribution, revenue contribution by customer state, and holiday versus non-holiday revenue. A second dashboard page summarizes the comparison between ETL and ELT approaches.

VIII. ETL AND ELT COMPARISON

The ETL and ELT pipelines use the same data sources and produce consistent final fact table row counts. The main difference is located in the transformation stage. In ETL, transformations are performed before loading data into the warehouse using Python and Pandas. In ELT, raw data is loaded first into the warehouse, and transformations are performed afterward using SQL.

The ETL approach is suitable when data cleaning and preparation must be controlled before loading into the warehouse. The ELT approach is suitable when raw data needs to be preserved in the warehouse and transformations can be executed closer to the analytical storage layer. In this project, both approaches successfully produce a final star schema and support the same analytical needs.

IX. CONCLUSION

This project successfully implements ETL and ELT pipelines using Olist e-commerce data and Nager.Date Holiday API data. Both pipelines produce SQLite-based star schema data warehouses with a final fact table containing 112,650 rows. The analytical SQL queries and Power BI dashboard support business insight exploration related to revenue, orders, products, payments, customer regions, delivery performance, reviews, and holiday effects. The comparison shows that ETL and ELT differ mainly in the timing and location of transformation while producing consistent final analytical outputs.

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